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BrightLearn

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Advanced SQL (NULL functions)

Question 1 →

```
SELECT employee-Id,  
       name,  
       COALESCE(salary, 0) AS salary-with-default  
FROM Employees
```

↓ outcome

employee-Id	name	salary-with-default
1	Alice	5000
2	Bob	0
3	charlie	7000
4	Dana	0

Question 2 →

```
SELECT  
    employee-id,  
    name,
```

```
    COALESCE(department, 'Not Assigned') AS  
    department-name
```

```
FROM Employees;
```

↓ outcome

employee-id	name	department-name
1	Alice	HR
2	Bob	IT
3	Charlie	Not Assigned
4	Dana	Finance

Question 3 →

```
SELECT  
    order-id,  
    customer-id
```

```
FROM Orders
```

```
WHERE customer-id IS NULL;
```

↓ outcome

order-id	customer-id
103	NULL

Question 4

```
SELECT
    order-id,
    customer-id,
    COALESCE(CAST (delivery-date AS STRING), 'Pending')
    AS delivery-status
FROM orders;
```

↓ outcome

order-id	customer-id	delivery-status
101	201	2024-12-01
102	202	Pending
103	NULL	2024-12-03

Question 5

```
SELECT
    student-id,
    name,
    COALESCE(grade, 0) AS final-grade
FROM students;
```

↓ outcome

student-id	name	final grade
1	Ethan	85
2	Maya	0
3	Olivia	90

Question 6

```
SELECT  
    COUNT(*) AS not-graded-count  
FROM students  
WHERE grade IS NULL;
```

↓ outcome

not-graded-count
1

Question 7

```
SELECT  
    product_id,  
    name,  
    price - COALESCE(discount, 0) AS final-price  
FROM Products;
```

↓ outcome

product_id	name	final-price
501	Keyboard	25
502	Mouse	10
503	Monitor	100

Question 8

```
SELECT  
COUNT(*) AS missing_email_count
```

```
FROM customers  
WHERE email IS NULL;
```

↓ outcome

missing_email_count
2

Question 9

```
SELECT  
customer_id,  
name,
```

```
COALESCE(email, 'No Email') AS email_display
```

```
FROM customers;
```

↓ outcome

Customer-id	name	email_display
1	Linda	No Email
2	Joseph	joseph@mail.com
3	Nia	No Email

Question 10

```
SELECT payment-id,  
COALESCE(method, 'Unknown') AS method-display,  
status  
FROM payments;
```



payment-id	method-display	status
301	Credit	NULL
302	Paypal	Success
303	Unknown	Failed

Question 11

Item-id	Item-name	quantity-checked
1	Pen	0
2	Notebook	150
3	Eraser	0

↑ outcome

```
SELECT item-id,  
item-name,  
COALESCE(quantity, 0) AS quantity-checked  
FROM Inventory;
```

Question 12

SELECT emp-id,

COALESCE(bonus, commission) AS first-available-reward

FROM Employees-Extra;



emp-id	first-available-reward
1	300
2	100
3	NULL

Question 13

SELECT
COUNT(*) AS no-room-count

FROM classes

WHERE room IS NULL;



no-room-count
2

Question 14

```
SELECT student-id,  
       date,  
       COALESCE(status, 'Not Marked') AS attendance-status  
FROM Attendance;
```

↓ outcome

Student-id	date	attendance-status
1	2025-04-01	Not Marked
2	2025-04-01	Present
3	2025-04-01	Present

Question 15

```
SELECT account-id,  
       COALESCE(account-type, 'Unknown') AS type-display,  
       COALESCE(balance, 0) AS balance-checked  
FROM Bank-Accounts;
```

↓ outcome

account-id	type-display	balance-checked
A1	Savings	0 5000
A2	Current	
A3	Unknown	2000

Question 16 →

```
SELECT project-id,  
       title,
```

```
       COALESCE(start-date, 'TBD') AS start-display  
FROM Projects;
```

↓ outcome

project-id	title	Start-display
1	Website Revamp	2025-01-10
2	Mobile App	TBD
3	Data Migration	TBD

Question 17 →

```
SELECT review-id,  
       product-id,
```

```
       COALESCE(comment, 'No Comment') AS comment-display
```

```
       COALESCE(rating, 0) AS rating-display
```

```
FROM Reviews;
```

↓ outcome

review-id	product-id	comment-display	rating-display
1	501	Great product	4
2	502	No Comment	0
3	503	Works fine	3

Question 18

SELECT supplier-id,
name,

COALESCE (phone, alt-phone, 'No Contact') AS
contact-number

FROM suppliers;

↓ outcome

Supplier-id	name	Contact-number
1	Global Goods	123456789
2	Best Supplies	987654321
3	ValueSource	No Contact

Question 19

SELECT user-id

COALESCE (theme, 'Light') AS theme-set,

COALESCE (language, 'English') AS language-set,

COALESCE (timezone, 'UTC') AS timezone-set

FROM User-Settings;

↓ outcome

user-id	theme-set	language-set	timezone-set
1	Light	English	UTC
2	Dark	English	UTC+1
3	Light	English	UTC

Question 20

```
SELECT record-id,  
       machine-id,  
       COALESCE(issue, 'Unknown Issue') AS issue-log,  
       COALESCE(technician, 'Not Assigned') AS technician-name  
FROM maintenance;
```

↓ outcome

record-id	machine-id	issue-log	technician-name
1	M101	Overheating	Not Assigned
2	M102	Unknown Issue	Not Assigned
3	M103	Jammed	Alex

THE END